

On a Claimed Upper Bound for Partial Desarguesian Parallelisms

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Abstract

Johnson claimed that, when t is prime, a partial Desarguesian t -parallelism in $V(zt, q)$ has at most

$$\left\lfloor \frac{q^{zt-1} - 1}{q^{t-1} - 1} \right\rfloor$$

members, and concluded that no Desarguesian t -parallelism exists for odd prime t . We show that this bound fails throughout its nontrivial odd-prime parameter range. The Cayley-graph construction of Zhang and Zhou uses $\text{GL}(zt, q)$ -images of a standard Desarguesian spread. These images remain Desarguesian, and their lower bound is strictly larger than Johnson's claimed upper bound for every prime power q , odd prime $t \geq 3$, and $z \geq 2$. The comparison is elementary once the two bounds are placed in the same notation.

We also locate the unsupported step in the proposed correspondence underlying the bound: affine lines arising from two potentially different extension-field structures are treated as a common line merely because their defining points belong to the set-theoretic intersection. Johnson's later book itself records the associated disjointness issue as unresolved while retaining the claimed theorem. Earlier work on $PG(5, 2)$ already implicitly supplies a complete finite counterexample, so no priority is claimed for the first counterexample. A compact certificate of 18 pairwise disjoint Desarguesian 3-spreads in $V(6, 2)$ is included as an independently checkable finite witness.

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1 Introduction

Let $V = V(n, q)$ be an n -dimensional vector space over $\mathbb{F}_q$, and suppose that $t \mid n$. A t -spread of V is a partition of the nonzero vectors into t -dimensional $\mathbb{F}_q$ -subspaces. It is Desarguesian when V can be regarded as a vector space over $\mathbb{F}_{q^t}$ and the spread members are the one-dimensional $\mathbb{F}_{q^t}$ -subspaces. A partial t -parallelism is a family of t -spreads no two of which share a t -dimensional subspace.

For $n = zt$, write $D(n, t, q)$ for the largest size of a partial t -parallelism all of whose spreads are Desarguesian. Johnson [1, Corollary 1] claimed, for prime t , the upper bound

$$D(zt, t, q) \leq J(zt, t, q) := \left\lfloor \frac{q^{zt-1} - 1}{q^{t-1} - 1} \right\rfloor. \quad (1)$$

The paper then used (1) to rule out Desarguesian t -parallelisms for odd prime t . An equivalent statement and bound reappear as Theorem 51 and Corollary 14 of Johnson's book [2, Chapter 10].

The purpose of this note is not to construct a new large family of spreads. Instead, it observes that a known construction already contradicts (1) in every nontrivial odd-prime case. Zhang and Zhou [6] derived lower bounds for partial t -parallelisms by taking an independent set in a Cayley graph on $\text{GL}(n, q)$. Their vertices act on a fixed field-reduction spread. Consequently, every spread in the resulting family is Desarguesian, even though their extremal notation does not impose that restriction.

Our main result is the following uniform correction.

Theorem 1.1. *Let q be a prime power, let $t \geq 3$ be an odd prime, let $z \geq 2$, and put $n = zt$. Then*

$$D(n, t, q) > \frac{q^t - 1}{q^n - 1} \left[\frac{n - 1}{t - 1} \right]_q > \frac{q^{n-1} - 1}{q^{t-1} - 1}. \quad (2)$$

In particular, $D(n, t, q) > J(n, t, q)$, so the upper bound (1) is false throughout this parameter range.

The first inequality in (2) comes from the Zhang–Zhou construction; the second is the new comparison needed to expose the contradiction.

Section 2 checks that the Zhang–Zhou witnesses satisfy Johnson's Desarguesian condition. Section 3 proves the strict comparison. Section 4 records the earlier finite counterexample and examines the proof step behind the claimed bound. Section 5 describes the independent finite certificate.

2 Desarguesianity of the orbit construction

Fix n divisible by t . A standard Desarguesian t -spread can be obtained by identifying $V(n, q)$ with $\mathbb{F}_{q^n}$ and taking

$$\mathcal{S}_0 = \{ \omega^i \mathbb{F}_{q^t} : 0 \leq i < (q^n - 1)/(q^t - 1) \}, \quad (3)$$

where ω is primitive in $\mathbb{F}_{q^n}$. Zhang and Zhou [6, Theorem 3.1] form a Cayley graph on $\text{GL}(n, q)$ in which two matrices are adjacent precisely when the corresponding images of $\mathcal{S}_0$ share a spread member. An independent set therefore produces a partial t -parallelism consisting of spreads $M\mathcal{S}_0$.

The following elementary observation places this construction inside the restricted class counted by $D(n, t, q)$.

Lemma 2.1. *For every $M \in \text{GL}(n, q)$, the spread $M\mathcal{S}_0$ is Desarguesian.*

Proof. Let

$$\rho : \mathbb{F}_{q^t} \longrightarrow \text{End}_{\mathbb{F}_q}(V)$$

be the extension-field embedding whose one-dimensional orbits give $\mathcal{S}_0$. Conjugation defines another embedding

$$\rho_M(a) = M\rho(a)M^{-1}.$$

The one-dimensional $\mathbb{F}_{q^t}$ -subspaces for the extension structure ρ_M are exactly the M -images of those for ρ . Hence $M\mathcal{S}_0$ is Desarguesian. $\square$

Johnson's definition requires each spread in the partial parallelism to be Desarguesian; it does not require all spreads to arise from one common extension-field embedding. Lemma 2.1 is therefore sufficient.

Zhang and Zhou prove [6, Corollary 3.4] that their construction contains more than

$$L(n, t, q) := \frac{q^t - 1}{q^n - 1} \left[\begin{matrix} n-1 \\ t-1 \end{matrix} \right]_q \quad (4)$$

spreads. Since the witnesses used in that proof are the orbit images just described, Lemma 2.1 gives

$$D(n, t, q) > L(n, t, q). \quad (5)$$

3 Uniform comparison with the claimed upper bound

Put

$$J_0(n, t, q) = \frac{q^{n-1} - 1}{q^{t-1} - 1},$$

so that $J(n, t, q) = \lfloor J_0(n, t, q) \rfloor$. Expanding the Gaussian binomial coefficient in (4) and cancelling the factor with index zero gives

$$\frac{L(n, t, q)}{J_0(n, t, q)} = \frac{q^t - 1}{q^n - 1} \prod_{i=1}^{t-2} \frac{q^{n-1-i} - 1}{q^{t-1-i} - 1}. \quad (6)$$

We prove that this ratio is greater than one.

Lemma 3.1. *If $t = 3$ and $n = 3z$ with $z \geq 2$, then $L(n, 3, q) > J_0(n, 3, q)$.*

Proof. Equation (6) becomes

$$\frac{L(n, 3, q)}{J_0(n, 3, q)} = \frac{(q^2 + q + 1)(q^{n-2} - 1)}{q^n - 1}.$$

The numerator minus the denominator is

$$(q^2 + q + 1)(q^{n-2} - 1) - (q^n - 1) = (q + 1)(q^{n-2} - q),$$

which is positive because $n \geq 6$. □

Lemma 3.2. *If $t \geq 4$, $n = zt$, and $z \geq 2$, then $L(n, t, q) > J_0(n, t, q)$.*

Proof. For $1 \leq i \leq t-2$,

$$\frac{q^{n-1-i} - 1}{q^{t-1-i} - 1} > q^{n-t}, \quad (7)$$

because the numerator of the left side exceeds $q^{n-t}(q^{t-1-i} - 1)$. Also,

$$\frac{q^t - 1}{q^n - 1} = q^{t-n} \frac{1 - q^{-t}}{1 - q^{-n}} > (1 - q^{-t})q^{t-n}. \quad (8)$$

Substituting (7) and (8) into (6) yields

$$\frac{L(n, t, q)}{J_0(n, t, q)} > (1 - q^{-t})q^{(t-3)(n-t)}. \quad (9)$$

For $t = 4$, the right side is minimized at $q = 2$ and $n - t = 4$, where it is

$$\frac{15}{16} 2^4 = 15 > 1.$$

For $t \geq 5$, it is minimized at the boundary $q = 2$, $t = 5$, and $n - t = 5$, where it is

$$\frac{31}{32} 2^{10} > 1.$$

Thus the ratio in (6) is greater than one. □

Proof of Theorem 1.1. By Lemma 2.1 and the Zhang–Zhou construction, (5) holds. Lemma 3.1 handles $t = 3$, and Lemma 3.2 handles every odd prime $t \geq 5$. Therefore

$$D(n, t, q) > L(n, t, q) > J_0(n, t, q) \geq \lfloor J_0(n, t, q) \rfloor,$$

which is (2). □

The argument actually compares the two numerical bounds whenever $t \geq 3$, $t \mid n$, and $n \geq 2t$; primality is retained in Theorem 1.1 because that is the scope of Johnson’s claim under correction.

4 Historical boundary and the failed proof bridge

4.1 An earlier complete finite counterexample

The parameter triple $(q, t, z) = (2, 3, 2)$ already shows why the present note cannot claim the first counterexample. Projective planes in $PG(5, 2)$ correspond to three-dimensional vector subspaces of $V(6, 2)$. Topalova and Zhelezova [4] proved that $PG(5, 2)$ has only one equivalence class of projective 2-spreads under $PGL(6, 2)$. A standard field-reduction spread from $\mathbb{F}_{64}/\mathbb{F}_8$ is Desarguesian. Since $PGL(6, 2) = GL(6, 2)$, every 2-spread in this geometry is therefore Desarguesian.

Sarmiento [3] had already constructed resolutions of the 2-(63, 7, 15) design associated with $PG(5, 2)$; these resolutions are complete projective 2-parallelisms. The number of projective planes is

$$\begin{bmatrix} 6 \\ 3 \end{bmatrix}_2 = 1395,$$

and every spread contains

$$\frac{2^6 - 1}{2^3 - 1} = 9$$

planes. A complete parallelism consequently contains $1395/9 = 155$ spreads, all Desarguesian by the uniqueness result. Johnson’s claimed bound at these parameters is only

$$\left\lfloor \frac{2^5 - 1}{2^2 - 1} \right\rfloor = 10.$$

Thus the older literature already implicitly contains the contradiction $155 > 10$. Later work of Topalova and Zhelezova [5] gives further explicit constructions and classifications in the same geometry.

4.2 Where the proposed correspondence loses a hypothesis

Johnson derives (1) from a correspondence between partial Desarguesian t -parallelisms and translation nets assembled from rational Desarguesian partial spreads [1, Theorem 3]. In the proof, two affine planes π_1, π_2 arise from two Desarguesian spreads. These spreads may use different copies of $\mathbb{F}_{q^t}$ inside $\text{End}_{\mathbb{F}_q}(V)$.

For two points $P, Q \in \pi_1 \cap \pi_2$, the proof then treats the affine line through P and Q as a line common to both planes. Uniqueness of a line through two points holds within each affine-plane incidence structure, but it does not identify the two lines when the extension-field structures differ. The same issue occurs when 0 and $Q - P$ are used to infer a common component. Consequently, the set-theoretic intersection $\pi_1 \cap \pi_2$ has not been shown to be an affine subplane of either plane, and the subsequent subfield dimension argument lacks its required premise.

The book version makes the status of this bridge especially clear. Before restating the result, Johnson [2, pp. 134–136] says that it is “not clear” whether the two rational Pappian partial spreads are disjoint outside their common regulus. Theorem 51 nevertheless restates the equivalence, leaves the common-affine-subplane assertion to the reader, and Corollary 14 repeats the upper bound.

The numerical contradiction in Theorem 1.1 proves that this unresolved bridge cannot hold under the stated hypotheses in general. This note does not attempt to classify additional compatibility assumptions under which a modified correspondence might be valid.

5 A compact independent certificate

The accompanying verification data give 18 explicit matrices in $\text{GL}(6, 2)$. Starting from the field-reduction spread $\mathbb{F}_{64}/\mathbb{F}_8$, the checker reconstructs the 18 image spreads and verifies that

1. every matrix is invertible;
2. every image is a 3-spread of $V(6, 2)$;
3. the 18 spreads are distinct; and
4. every pair is disjoint as a set of three-dimensional subspaces.

Every image is Desarguesian by Lemma 2.1, so the certificate independently establishes

$$D(6, 3, 2) \geq 18 > 10.$$

It is intentionally smaller than the old complete 155-spread witness. Its purpose is reproducibility: checking the correction does not require reconstructing the earlier full classifications or parallelisms. The checker also contains stable destructive controls which reject a singular map and an intentionally duplicated spread.

From this public entry directory, run

```
python -X utf8 verification\check_desarguesian_certificate.py
```

The finite certificate is supplementary evidence. The general result in Theorem 1.1 rests on the symbolic comparison in Section 3, not on finite enumeration.

6 Scope of the correction

The mathematical contribution of this note is deliberately limited:

- it identifies the Zhang–Zhou orbit lower bound as a uniform contradiction to Johnson’s claimed upper bound over the whole nontrivial odd-prime range;
- it explains why all orbit witnesses meet Johnson’s individual Desarguesianity requirement;
- it shows that the unresolved common-subplane bridge is fatal to the claimed theorem under its published hypotheses; and
- it supplies a compact independent finite witness.

The note does not claim the first finite counterexample, the first Desarguesian 3-parallelism, the first observation that the common-subplane step needs proof, or a best corrected replacement for Johnson’s correspondence. It also does not claim external expert confirmation or peer review.

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References

- [1] N. L. Johnson, *The non-existence of desarguesian t -parallelisms, t an odd prime*, Note di Matematica 30 (2010), no. 1, 13–16. <https://doi.org/10.1285/i15900932v30n1p13>
- [2] N. L. Johnson, *Combinatorics of Spreads and Parallelisms*, CRC Press, 2010, Chapter 10, Theorem 51 and Corollary 14. <https://doi.org/10.1201/EBK1439819463>
- [3] J. F. Sarmiento, *On point-cyclic resolutions of the 2 -(63,7,15) design associated with $PG(5,2)$* , Graphs and Combinatorics 18 (2002), 621–632. <https://doi.org/10.1007/s003730200046>
- [4] S. T. Topalova and S. D. Zhelezova, *Classification of the 2-spreads of $PG(5,2)$* , Proceedings of the Thirty Seventh Spring Conference of the Union of Bulgarian Mathematicians (2008), 163–168. https://www.math.bas.bg/smb/2008_PK/2008/pdf/163-168.pdf
- [5] S. Topalova and S. Zhelezova, *2-spreads and transitive and orthogonal 2-parallelisms of $PG(5,2)$* , Graphs and Combinatorics 26 (2010), 727–735. <https://doi.org/10.1007/s00373-010-0943-8>
- [6] T. Zhang and Y. Zhou, *New lower bounds for partial k -parallelisms*, Journal of Combinatorial Designs 28 (2020), 75–84. <https://doi.org/10.1002/jcd.21681>